

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

Sebastian Bobadilla-Suarez
Independent Researcher
`sebastian.bobadilla.s@gmail.com`
ORCID: 0000-0002-5951-0772

6 September 2026

Abstract

Coordinated cortical activity need not describe a single experienced situation. We develop a conditional field account of coherence, compatibility and self-representation. Declared actuator modes bound mean coupling by installed energy, giving a necessary resource condition for the scalar synchronization threshold. A shared scalar Lipschitz encoder converts phase coherence into a content-disagreement bound. Compatible spatial measures glue into one state; a contracting reconstruction map that fixes it makes it the unique fixed point. Approximate agreement and reconstruction give error bounds. Lean 4 checks these results and two compositions, each with eight connecting hypotheses and additional structures. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. The awakening hypothesis links extracellular geometry to increasing coupling. Finite ramps delay stationary onset, while coupling increase, diffusion decrease and a matched common drive can share stationary phase distributions. Testing requires calibrated physical identifications, resolved dynamics and independent measurements of content agreement and self-encoding.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Consider reaching for a cup. A visual description of its location and a description of the hand's relation to it can agree about where the cup is while contributing different contents. Synchronizing the two processes would leave disagreement about that shared quantity unresolved. An internal representation must additionally locate the perceiver within the represented situation. This example motivates the requirements; the mathematical constructions use spatial measures and scalar fields, whose identification with perceptual

23 content is a further modelling task.

24 The results specify both connections and separations between these re-
25 quirements. A declared actuator’s stored energy bounds its mean coupling,
26 giving a necessary energy condition for the scalar coherence threshold. Equal
27 phases permit incompatible local descriptions when content is independent
28 of phase; when a common Lipschitz encoder reads phase, coherence bounds
29 disagreement. Exact compatibility yields a unique global extension, and ap-
30 proximate compatibility bounds the variation among specified selections. A
31 contracting reconstruction map has a unique fixed point, with a quantita-
32 tive error bound, but identifying that point with the state formed from local
33 descriptions requires accuracy at that state.

34 Awakening gives the physical hypothesis a concrete test. We propose that
35 changing extracellular geometry increases coupling through the synchroniza-
36 tion threshold. The stationary onset is square-root, but finite ramp speeds
37 delay recovery. Moreover, increasing coupling, decreasing diffusion and a
38 matched common drive can produce identical stationary phase distributions.
39 A separate plasticity control preserves coherence while losing alignment with
40 its diagnostic environmental template. These results identify measurements
41 and mechanisms that a successful coordination model must supply.

42 The article follows the three requirements and their tests (Fig. 1). Ther-
43 modynamic constraints specify the costs of candidate implementations, in-
44 cluding agents that act on their future inputs. The detailed feedback, learn-
45 ing and resource models have distinct state spaces and assumptions; their

46 consequences do not together instantiate a cortical agent. Section 6 summa-
 47 rizes the conditional composition, and Supplemental Material provides the
 48 constructions, witnesses and numerical controls.

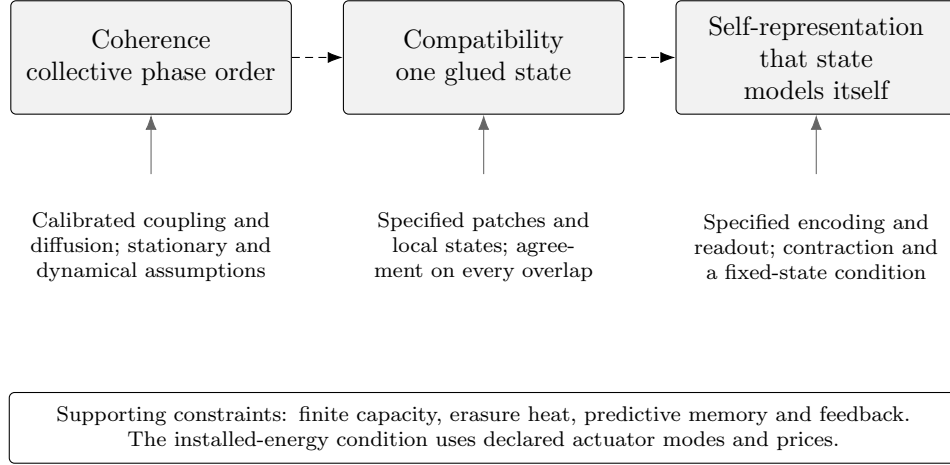


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

49 2 Coherence: physical coupling and its re- 50 source condition

51 2.1 Dynamics and the stationary threshold

52 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

53 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
54 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
55 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
56 parameters.

57 For identical frequencies in their rotating frame, the positive normalized
58 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

59 where I_n are modified Bessel functions and ψ is the collective phase. The for-
60 mal derivation uses the absence of stationary current for a periodic gradient
61 drift to obtain this density, then verifies normalization and self-consistency.
62 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one
63 positive solution coexists with zero and increases strictly with K . Its onset

64 is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

65 Figure 2 shows this stationary branch.

66 Linearization about the uniform density gives rate $K/2 - D$ for the first
67 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
68 exactly above threshold. About the coherent density, the quadratic form
69 is strictly dissipative transverse to collective rotation, whose rate is zero.
70 These linear estimates concern the limiting equation. Relating them to finite
71 stochastic trajectories requires a dynamical mean-field limit and branch selec-
72 tion; exchange symmetry alone does not establish the needed independence
73 (Sznitman, 1991).

74 **2.2 Installed energy bounds mean coupling**

75 A spatial model replaces the finite sum by integration against a kernel. One
76 declared implementation carries response modes with continuous profiles ϕ_i ,
77 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

78 The same configuration sets the reciprocal kernel and stored energy. An
79 executed transition changes the kernel through its changed occupancies and
80 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
81 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-

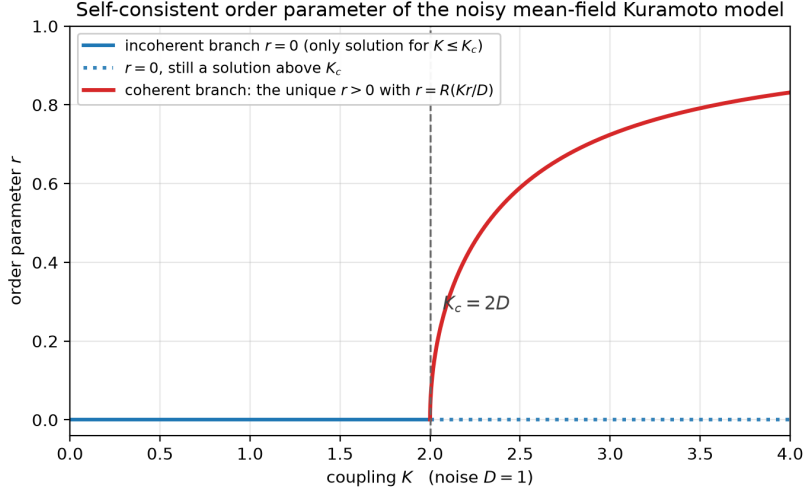


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

82 pair reconstruction on refining spatial partitions converges to its product-
 83 measure integral under the specified compactness and sampling assumptions;
 84 the refinement index is independent of the number of control updates.

85 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 86 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 87 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_z p(z) U(z), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned} \tag{5}$$

88 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 89 coherent stationary branch and any growing Fourier mode about the uniform
 90 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 91 is necessary: energy above it need not produce sufficient coupling because
 92 the bound can be loose.

93 The quantity constrained is installed energy. Dissipated heat does not
 94 determine it, and κ is supplied by the mode profiles, prices and substrate
 95 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 96 tial dynamics requires a justified reduction. Identifying the installed kernel
 97 with a cortical field is a separate physical commitment.

98 **2.3 Physical identification and calibration**

99 The cortical hypothesis identifies part of effective coupling with endogenous
 100 extracellular electromagnetic interactions. Fields can modulate spike timing
 101 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
 102 propose coordination of distributed neural information (Pinotsis & Miller,
 103 2022; McFadden, 2020). Phase organization under propofol motivates study-
 104 ing this regime (Bastos et al., 2021), but does not measure coupling.

105 The formal field predicate requires a jointly continuous kernel, a normal-
 106 ized substrate, coupling equal to the kernel average, and no single site pro-
 107 viding more than half that strength. Toy constructions satisfy it. A cortical
 108 realization must specify the scale at which the kernel and its reduction ap-
 109 ply: calcium imaging reports different state-dependent modularity patterns

110 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

111 A spike-time shift alone supplies no coupling rate. The dimensionless
112 product of population size N , timing sensitivity s , field amplitude E and
113 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (6)$$

114 This rate describes how induced timing shifts accumulate into drift. Nei-
115 ther it nor the installation factor κ follows from a heat bound. Testing the
116 physical threshold requires independent estimates of the interaction kernel,
117 its aggregation and diffusion in the same cortical regime.

118 **3 Testing coherence through recovery after** 119 **awakening**

120 The proposed control variable is extracellular geometry. Sleep-associated
121 changes in extracellular space (Xie et al., 2013) motivate coupling that evolves
122 more slowly than phase dynamics. Extracellular volume fraction is approxi-
123 mately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); the sleep
124 study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz,
125 cortical tissue conducts approximately resistively at $0.3\text{--}0.6 \text{ S m}^{-1}$ (Barbour,
126 2017). At fixed transmembrane current density, an effective-medium de-
127 scription predicts that a larger extracellular fraction raises conductivity and

128 reduces field amplitude.

129 The coupling-gated hypothesis consequently predicts that extracellular
130 contraction on awakening raises E and, holding the other factors in Eq. (6)
131 fixed, raises effective coupling. This argument supplies a direction; the
132 geometry–coupling relation, its magnitude and its role in human recovery
133 require measurement.

134 If $K(t)$ crosses K_c while the distribution tracks the coherent stationary
135 branch, Eq. (3) gives

$$r(t) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (7)$$

136 The crossing predicts latency only when coupling is measured independently
137 of coherence. Critical slowing down restricts branch tracking near threshold,
138 and finite populations fluctuate below it. An observable onset therefore needs
139 a criterion above that fluctuation floor.

140 **3.1 What a recovery distribution can identify**

141 Equation (2) gives a concentration–coherence relation with no fitted shape
142 parameter. Estimate r from circular resultant length and a from density
143 shape: inverting R from the measured r would make agreement automatic.
144 Even independently estimated agreement identifies a statistical family shared
145 by different mechanisms.

146 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-

147 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 148 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 149 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 150 and diffusion D_0 have that same density. Their stationary distributions co-
 151 incide window by window despite different causes.

152 Resolved dynamics supply additional information. In the ideal model with
 153 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 154 increment means constrain drift. Separating endogenous coupling from a
 155 matched common drive additionally requires a measured drive or controlled
 156 perturbation; identifying electromagnetic coupling requires evidence specific
 157 to that interaction. Supplemental Material gives the equivalence construction
 158 and its numerical check.

159 **3.2 Finite ramps change the observable onset**

160 The ramp study uses $N = 2000$, $D = 1$ and 32 replicas at four coupling
 161 speeds. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after
 162 threshold. Faster ramps delay escape (Fig. 3). Only 5/32 replicas escape
 163 within the fastest window. The three slower ramps have mean coupling
 164 delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents run from 0.898 to
 165 0.417, compared with 0.443 to 0.448 for the exact stationary branch on the
 166 same grids. At the fastest speed the ensemble mean never reaches the fit
 167 window; 0.964 is a local log-slope of a delayed trajectory.

168 This delayed passage through a noisy bifurcation (Berglund & Gentz,

2002) makes the apparent exponent depend on ramp speed and fitting window. A deviation from the square-root law tests the adiabatic approximation where branch tracking is independently justified. The sweep does not bracket the specified deviation-based critical ramp speed.

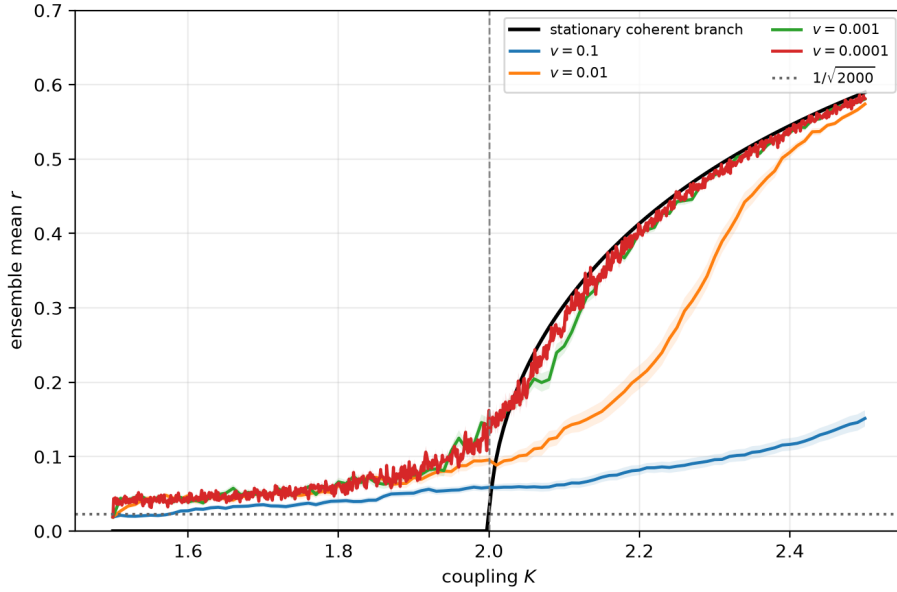


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in [1.958, 2.009]$ contains 2. These are finite numerical regimes, with full parameters and controls in Supplemental Material; they supply no cortical calibration.

178 **3.3 What the estimators produce on EEG**

179 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
180 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
181 itages produce different residuals. The bipolar results are compatible with the
182 Bessel relation, but their concentration range cannot distinguish it from its
183 linear approximation at the observed scatter. Temporal pooling changes the
184 estimand, and sedation recordings leave the awakening hypothesis untested.

185 An informative dataset needs broader concentration, calibrated estima-
186 tors and an observation model for dependence, source mixing and reference
187 choice. Public mouse calcium imaging provides thousands of spatially lo-
188 cated neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but
189 its 7.65 Hz frame rate is below the phase-estimation band used here. Site
190 count, state coverage and bandwidth are separate requirements.

191 **3.4 A protocol for the awakening experiment**

192 The physical test has six requirements:

- 193 1. **Coupling and diffusion in the same units.** Estimate them from re-
194 solved phase dynamics, separating heterogeneity and observation noise.
195 Measure the drive or perturb the system to distinguish endogenous cou-
196 pling.
- 197 2. **Geometry calibrated to coupling.** Validate the geometry observ-
198 able and its relation to effective coupling, with uncertainty. The con-

ductivity argument supplies only the expected direction.

3. **Adequate phase measurements.** The proposed spatial estimator uses at least 100 sites, resultant length for r , and unweighted regression of log density on centered cosine for a . Calibrate this numerical minimum under the actual dependence and concentration range.

4. **An independent recovery endpoint.** Measure behavioral recovery separately from phase organization.

5. **A predeclared onset criterion.** Set a coherence threshold above the finite-size floor and account for ramp speed.

6. **Matched model comparison.** Compare coupling increase, diffusion change, common drive and alternative recovery models under the same observation and filtering assumptions.

Failure of the concentration–coherence relation over an informative range constrains the density model. Failure of the calibrated geometry–coupling association constrains the physical gate. Better prediction of independent recovery by a competing mechanism counts against its proposed role. A favorable outcome still leaves the content of coordinated activity to be explained.

217 4 Compatibility: from coordinated activity 218 to a global description

219 4.1 Why common timing is insufficient

220 For overlapping patches U and V , compatibility means that restricting either
221 local description to $U \cap V$ gives the same object. Descriptions can agree on
222 shared quantities while contributing different information elsewhere. For the
223 spatial measures used here, compatible restrictions give a unique measure on
224 the covered domain: the gluing property formalized by a sheaf.

225 In the cup example, visual and reaching descriptions must express a
226 shared quantity, such as cup location, in common coordinates. Restriction
227 retains that quantity while dropping details particular to each description.
228 Agreement permits surface appearance and hand configuration to contribute
229 to one situation. The mathematical counterexample assigns the same phase
230 to every patch while two local mass profiles disagree on a shared site. Those
231 profiles have no common extension. Because they are chosen independently
232 of phase, synchrony places no constraint on their disagreement.

233 This distinction limits the biological interpretation. Receptive territories
234 and content domains must be related to spatial patches, and the descriptions
235 must support the required extension. Matching overlap marginals of de-
236 coded probability laws need not determine a unique global joint distribution
237 (Abramsky & Brandenburger, 2011). The spatial-measure result supplies a

precise construction; applying it to perceptual content requires a justified content model.

4.2 When coherence does constrain content

A declared relation between phase and content gives a positive result. Let each patch read one oscillator through the same scalar encoder e , evaluated on the circle point $(\cos \theta, \sin \theta)$. Assume e is L -Lipschitz in chord distance. Its observation is constant across that patch. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$ implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (8)$$

At $r = 1$, the observations agree exactly. The factor N makes this estimate potentially weak in large populations; it supplies a pointwise guarantee from a population average. The encoder and assignment of oscillators to patches are inputs. Arbitrary local decoders are outside this result.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i + \eta o_i$. Content and observation discrepancies bounded by ε and δ give a next discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (9)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-

256 duce an initial mismatch. Approximate observations leave a nondecaying
 257 term in the error bound. Taking a uniform bound on Eq. (8) connects this
 258 update to the phase process. These are mechanisms for specified scalar con-
 259 tents; their neural meaning and relevance to experience remain to be estab-
 260 lished.

261 **4.3 What approximate agreement determines**

262 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 263 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 264 Weights that vanish outside their patches and sum to one at each site select
 265 a global profile by weighted averaging. The selection lies within ε of each
 266 patch there, and any two such selections differ by at most ε , independently
 267 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 268 $L\varepsilon$ across selections.

269 At zero tolerance, the selection is the unique exact extension. At every
 270 positive tolerance there are families with no exact extension; selection then
 271 gives an approximation whose dependence on weights is bounded. Supple-
 272 mental Material gives a three-site witness attaining the bound. The content-
 273 dynamics estimate concerns real-valued fields; using it for nonnegative mass
 274 profiles additionally requires that those observations and updates preserve
 275 nonnegativity. Neither construction selects the neural cover or decoder.

276 4.4 Coherent adaptation can lose environmental align- 277 ment

278 An adaptive coupling rule might supply representational structure, but its
279 objective must reward the relevant agreement. For symmetric coupling, pair-
280 wise sine terms cancel in the population mean drift. Minimizing squared drift
281 at that fixed mean therefore rewards equalizing drift across nodes. In the
282 frequency-cluster control, this favors coupling between unlike-frequency clus-
283 ters, whereas the diagnostic environmental template rewards coupling within
284 clusters.

285 For fixed phases, the continuum objective is

$$\sigma(K) = \frac{1}{D} \int \left[\omega(x) + \int K(x, y) \sin(\theta_y - \theta_x) d\mu(y) \right]^2 d\mu(x). \quad (10)$$

286 The formalization constructs the drift operator A and adjoint, obtains $\nabla\sigma =$
287 $(2/D)A^*(AK + \omega)$, and proves descent for the specified gradient flow. This
288 functional differs from thermodynamic entropy production: at a coherent sta-
289 tionary density the probability current vanishes while phase-averaged squared
290 drift is positive.

291 The finite control co-evolves noisy phases and symmetric, nonnegative
292 couplings at fixed total resource. Its template is a diagnostic covariance of
293 latent signals, never a training input. Across three seeds, second-half order
294 remains between 0.9167 and 0.9222, while the within-cluster to between-
295 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).

296 Random-update, relabelled-gradient and frozen-kernel controls are reported
297 in Supplemental Material.

298 The immediate post-update objective reduction disappears when aver-
299 aged over complete update intervals. A prespecified grid of 6 rate-cadence
300 combinations contains no case meeting the 5% sustained-reduction criterion
301 while retaining coherence and losing alignment. The finding is coherent adap-
302 tation away from the specified structure, with no demonstrated sustained cost
303 reduction.

304 Kernel alignment and overlap compatibility are different observables.
305 This control addresses acquisition of its designated template under its par-
306 ticular objective. Learning models can explicitly include data in an objective
307 (Corberi et al., 2026); an account of agreement about the cup must specify
308 the content its learning rule is meant to preserve.

309 **4.5 What must be specified and measured**

310 A neural test needs independently identified domains, local descriptions and
311 restriction maps. Two decoders should estimate a shared content on over-
312 lapping domains before their agreement is examined. Discrepancy requires
313 a null under decoding noise and controls using deliberately compatible and
314 incompatible examples. Shared priors can manufacture agreement, while
315 regional bias can manufacture disagreement. A phase-agreement estimator
316 would measure the variable whose explanatory limits are at issue.

317 Choosing the cover also chooses a candidate subject: trivial descriptions

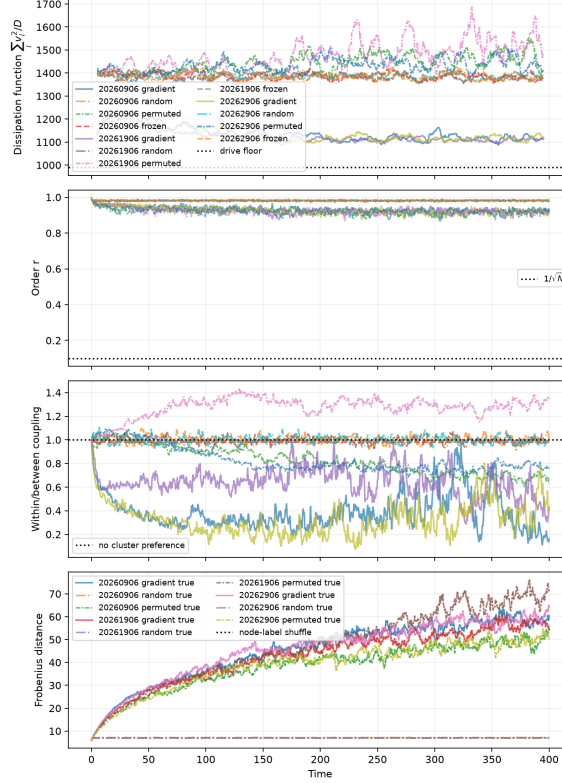


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The scalar mechanisms above identify additional hypotheses under which agreement can arise; they leave the empirical choice of content and the connection to that cover explicit.

5 Self-representation: a state that models itself

5.1 The reflexive construction

Compatibility gives a global state. Reflexivity relates it to an encoding within that state. Equip global sections with a complete metric. If the additional restriction-resonance condition identifies encoding with restriction to a region U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (11)$$

The region and readout are fixed across the state space. If F contracts and fixes the section s_* formed from the local descriptions, Banach's theorem makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.

Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than one for $\tau > 0$. Constructing a map with that factor remains a modelling obligation.

In the cup example, the global state includes the represented scene and the perceiver's relation to it. A local encoding must retain enough information for the readout to recover that modelled state; cup position alone would be insufficient if hand configuration and viewpoint could not be reconstructed. Here *prediction* names a reconstruction map. Interpreting it as temporal neural prediction requires dynamics.

For the same map with Lipschitz factor $q < 1$, approximate reconstruction gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (12)$$

At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens. A biological interpretation must specify the represented variables, metric and acceptable fidelity; changing readouts need a further model.

5.2 Why connect these properties to unity and a minimal self?

We propose the state glued from compatible local descriptions as a correlate of conscious unity, and its accurate internal reconstruction as a correlate of a minimal self. The motivations are that different contents belong to one ex-

355 perience situation, and that the situation is experienced from a perspective.

356 Compatibility expresses agreement about shared quantities while allow-
357 ing different contributions. It also permits a description of uncertainty or
358 conflicting judgments: apparent cup position and judged cup position can
359 be represented as distinct quantities within a consistent account of their dis-
360 crepancy. Defining which quantities are shared is substantive. A choice made
361 solely to secure agreement would deprive the proposal of empirical force.

362 The minimal-self proposal concerns the perceiver’s position and condition
363 in that situation: where the cup appears relative to a body that can reach it.
364 Such an encoding could be implicit and need carry no autobiographical nar-
365 rative. Its proposed necessity is philosophical: even an unreflective episode
366 has a perspective. The gluing theorem alone requires no self-encoding.

367 Reconstruction also needs a functional interpretation. A reconstructable
368 recording need never be used to distinguish changes in a system from changes
369 in its surroundings. Altering the encoded hand or viewpoint while holding
370 the external scene fixed should change estimates of the perceiver’s relation
371 to the cup as the readout predicts. Whether that represented perspective is
372 experienced is a further question.

373 **5.3 Selecting and testing the construction**

374 Both uniqueness results concern specified objects. They choose neither the
375 cover nor the encoding, and do not select between candidate subjects. Inte-
376 grated information theory treats exclusion explicitly (Tononi et al., 2016); the

377 present construction leaves selection to independent evidence about neural
378 organization.

379 A constant readout returning a predetermined state contracts and has a
380 fixed point while discarding its encoding. It fails the functional sensitivity
381 required above. Local descriptions must predict the contents assigned to
382 their domains; encoding and readout must be specified before testing their
383 responses over a declared range.

384 A test of the full proposal must separately measure coupling relative to
385 diffusion, content compatibility and self-reconstruction, alongside an inde-
386 pendent endpoint for experience. Meeting all three without a unified episode
387 would challenge sufficiency. A unified episode with incompatible descrip-
388 tions would challenge the unity identification; one lacking the specified self-
389 encoding would challenge the reflexive requirement. Awakening, dreaming
390 and dissociative states offer possible settings, but absence of report cannot
391 by itself define absence of experience.

392 The compatibility observable is currently a specification, and the neural
393 self-encoding has yet to be identified. Gluing and fixed-point conditions
394 permit multiple substrates; the electromagnetic commitment enters through
395 physical calibration. The proposal concerns which state is conscious and
396 leaves unexplained why those relations would be accompanied by experience.

397 6 Thermodynamics and composition

398 The phase and representation results constrain a candidate implementation
399 through different quantities: coupling, content disagreement and reconstruc-
400 tion error. Thermodynamics additionally constrains its physical operations.
401 Finite-register capacity obeys $H(p) \leq \log |X|$; erasure heat depends on
402 specified system–bath assumptions (Landauer, 1961; Norton, 2005; Shenker,
403 2000). A separate boundary result concerns a field reaching disconnected
404 vacuum components (Kibble, 1976). Their physical connections are inputs
405 to the composition below.

406 6.1 Feedback changes the information constraint

407 For passive memory X_t driven by an environmental signal S_t , the predictive
408 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (13)$$

409 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
410 et al., 2012). Zero retained information also satisfies it, so a task and a
411 learning rule require additional specification. Relating this bound to Lan-
412 dauer heat requires a common thermal scale and budget, and memory waste
413 bounded by erased entropy.

414 An acting agent changes the signal it will receive. Specify policy, world,

415 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{14}$$

416 The second conditional term vanishes in the passive model. With feedback,
417 the signed difference can be negative as action creates correlations with the
418 future. A causal reading requires an environmental state sufficient to describe
419 the relevant dynamics.

420 For a controlled substep, the active thermodynamic bound instead uses
421 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
422 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
423 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
424 2013). Every forward path of positive probability must have a positive-
425 probability reverse for this finite identity. A missing reverse gives infinite
426 divergence. The supplement develops the support conditions and composes
427 actuation with memory updates on their shared intermediate law.

428 **6.2 Learning and repeated operation need explicit re-** 429 **sources**

430 The finite examples distinguish selecting a feasible policy, relaxing a register
431 toward an encoded reward, and learning an unknown task from observa-
432 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (15)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply automatically.

450 6.3 What the composition establishes

451 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 452 cept eight connecting implications (Table 1). In the active branch, a named
 453 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (16)$$

454 with B the register’s heat. The upper allocation follows from a ledger whose
 455 operations deliver that total heat to one reservoir and whose other operations
 456 do not increase joint entropy. Those ledger assumptions are physical inputs.
 457 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A separate active scalar construction uses the process’s entropy change with a supplied gain, profile and regular meshes.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, the installed kernel can meet E56 and E67 only if $U_{\text{inst}} > 2D/\kappa$. This necessary condition supplies neither physical premise.

Both branches also take a thermodynamic cover carrying exact over-
lap compatibility and a reflexive boundary specifying encoding and readout.
They conclude **UnifiedSelf**: the glued section is the unique fixed point of
self-prediction. Compatibility and accuracy at that section enter as premises;
uniqueness follows. Identifying encoding with restriction needs the separate
resonance condition of Eq. (11).

Joint witnesses combine a register, boundary, refining mesh and small
spatial cover; active witnesses add controlled thermal updates and a priced
installation. They establish satisfiability under their stated identifications.

467 Mesh convergence follows from refinement data, and the witnesses establish
468 no common cortical mechanism. Supplemental Material maps the claims to
469 their Lean declarations and records the counterexamples. The development
470 declares no additional axioms beyond its libraries' foundational axioms.

471 **7 Discussion**

472 The framework distinguishes the physical conditions for coherent activity
473 from the relations needed to interpret it as one internally represented sit-
474 uation. Its positive connections depend on specified mechanisms: installed
475 modes relate energy to mean coupling, a shared phase encoder relates coher-
476 ence to scalar content agreement, and an observation update propagates that
477 agreement into retained fields. Their assumptions identify what a cortical
478 model must establish.

479 The separation results delimit what would count as evidence. A station-
480 ary recovery distribution leaves coupling, diffusion and a matched common
481 drive indistinguishable. Synchronous patches can carry incompatible phase-
482 independent descriptions. Coherent adaptation under the tested squared-
483 drift objective can lose environmental alignment. Each result concerns the
484 stated model and observable; none determines cortical learning from ther-
485 modynamics or diagnoses experience from synchrony alone.

486 For the awakening hypothesis, the next requirement is a calibrated
487 geometry-coupling relation and resolved phase dynamics, with an indepen-

488 dent recovery endpoint. The installation bound adds a necessary resource
489 condition where its hardware representation applies. It does not calibrate
490 cortical mode prices or replace evidence that the physical field operates
491 above threshold.

492 For the representational connection, the informative experiment is
493 whether a controlled change in coordination alters independently decoded
494 content agreement. Decoders and restriction maps must first recover compat-
495 ible and incompatible cases under realistic noise. Perturbations then need
496 controls for sensory information and decoder quality. The phase-encoder
497 theorem supplies one mechanism to test, but its scalar observations and
498 population-size dependence limit how directly it can describe neural content.

499 The self clause requires a candidate encoding whose perturbation changes
500 the represented scene, body or viewpoint as its readout predicts. Approx-
501 imate gluing and reconstruction make fidelity an explicit parameter. They
502 leave the cover, metric, encoder and required resolution to be justified inde-
503 pendently; uniqueness within these choices does not choose a subject.

504 Integrated-information, global-workspace and predictive-processing ac-
505 counts organize unity, access and modelling differently (Tononi et al., 2016;
506 Baars, 1997; Dehaene et al., 2014; Friston, 2010). The present proposal con-
507 tributes a candidate physical interaction and conditional relations between
508 local content and self-encoding. Its explanatory value depends on identifying
509 those relations in neural activity and showing that they matter for experi-
510 ence. In the cup example, the challenge is to explain how coordination helps

511 maintain one represented situation and how the perceiver’s place within it
512 contributes to the episode.

513 **Data and code availability**

514 The Lean development, simulation scripts, tests and saved numerical sum-
515 maries are maintained in the manuscript source repository. Publication sim-
516 ulation values are regenerated from saved summaries without rerunning pro-
517 duction sweeps. The exploratory EEG analysis uses OpenNeuro dataset
518 ds005620 (Bajwa et al., 2025); preprocessing and selection are described
519 in Supplemental Material. The source repository contains the reproducible
520 analysis and does not redistribute the raw recordings.

521 **Funding and competing interests**

522 This research received no external funding. The author declares no compet-
523 ing interests.

524 **Use of AI tools**

525 AI assistance to this work was substantial and is not separable by task.
526 Several coding assistants and chat interfaces were used in an interleaved way
527 throughout: Google’s Antigravity CLI (Gemini 3.1, Claude 4.6 Opus, Claude
528 4.6 Sonnet), OpenAI’s Codex CLI (Astra, Terra, Luna, Sol), Claude Code

529 (Claude Opus 5), Gemini Pro through its web interface, and Fable (Claude
 530 Opus 4) through OpenRouter, accessed with the Hermes CLI. Between them
 531 they contributed to the Lean development, the simulation and analysis code,
 532 and the drafting and revision of both documents. Because the tools were
 533 used in combination and in alternation, no section, proof, figure or result is
 534 attributable to any one of them, and none is claimed to be. The author is
 535 responsible for all content, including everything a tool produced. Machine
 536 checking establishes the stated Lean results under their hypotheses; it does
 537 not validate biological interpretations or replace scientific review of generated
 538 code and prose.

539 **References**

- 540 Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable
 541 electric fields emerge from and allow representational drift. *NeuroImage*,
 542 253, 119058.
- 543 Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated informa-
 544 tion theory: from consciousness to its physical substrate. *Nature Reviews*
 545 *Neuroscience*, 17(7), 450-461.
- 546 Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the
 547 Mind. *Oxford University Press*.
- 548 Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a compu-

549 tational theory of conscious processing. *Current Opinion in Neurobiology*,
550 25, 76-84.

551 Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature*
552 *Reviews Neuroscience*, 11(2), 127-138.

553 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
554 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
555 40, 24-50.

556 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
557 parts of the free energy principle. *Entropy*, 23(3), 293.

558 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
559 peror's New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.

560 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
561 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.

562 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
563 Electric fields sculpt neural activity and "tune" the brain's infrastructure.
564 *Progress in Neurobiology*, 226, 102465.

565 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
566 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
567 9(8), 45-60.

568 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
569 testable hypothesis about the characteristics of conscious as opposed to
570 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.

571 McFadden, J. (2020). Integrating information in the brain's EM field: the
572 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
573 niaa016.

574 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
575 guide neocortical network activity. *Neuron*, 67(1), 129-143.

576 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
577 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
578 95-103.

579 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
580 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.

581 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
582 on brain circuits in rats and humans. *Nature Communications*, 9, 483.

583 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
584 *logical Reviews*, 98(1), 239-389.

585 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
586 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
587 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.

588 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
589 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
590 ity of complexity measures to capture dreaming during propofol sedation.
591 *Scientific Reports*, 15, 32746.

592 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiagarajan, M., ... &
593 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
594 brain. *Science*, 342(6156), 373-377.

595 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
596 *Physiological Reviews*, 88(4), 1277-1340.

597 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
598 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

599 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
600 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
601 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

602 Landauer, R. (1961). Irreversibility and heat generation in the computing
603 process. *IBM Journal of Research and Development*, 5(3), 183-191.

604 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
605 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

606 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
607 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

608 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
609 of prediction. *Physical Review Letters*, 109(12), 120604.

610 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
611 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

612 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
613 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
614 031015.

615 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
616 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
617 PhysRevLett.111.180603.

618 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
619 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.

620 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
621 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
622 181.

623 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
624 archive.pitt.edu/115.

625 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
626 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
627 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.

- 628 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
629 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
630 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 631 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
632 bifurcations with additive noise. *Probability Theory and Related Fields*,
633 122, 341–388. doi:10.1007/s004400100174.
- 634 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
635 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
636 doi:10.1088/1367-2630/13/11/113036.
- 637 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
638 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave
639 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
640 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.
- 641 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
642 terns underlie complex brain dynamics and are related to cognitive pro-
643 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
644 023-01626-5.
- 645 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
646 and dissipation with partially observed fluctuating signals. *Physical Review*
647 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

- 648 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
649 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
650 cell resolution functional networks during unconsciousness are segre-
651 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
652 doi:10.1016/j.celrep.2025.116902.
- 653 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
654 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
655 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.